# Pulmonary Circulation

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Learning Objectives (5)

*After completing this brick, you will be able to:*

1 Compare and contrast systemic and pulmonary circulations.

2 Describe the distribution of pulmonary blood flow and the concept of lung zones.

3 Explain how pulmonary vascular resistance changes in response to cardiac output, arterial pressure, and lung volume variations.

4 Discuss the effects of hypoxic pulmonary vasoconstriction on blood flow distribution in the lungs.

5 Identify the mechanisms and causes of pulmonary edema.

**CASE CONNECTION**

CH, a 65-year-old man presents to the pulmonary clinic with progressive dyspnea. He has a history of chronic obstructive pulmonary disease (COPD) and worsening oxygen saturation over several weeks. During his workup, his physician notes signs of pulmonary hypertension and considers the impact of altered pulmonary blood flow.

As you read this brick, consider how you would explain the pulmonary circulatory changes in CH's condition, particularly concerning hypoxic vasoconstriction and pulmonary vascular resistance.

GO TO CONCLUSION ↓

## Compare and Contrast Systemic and Pulmonary Circulations (LO1)

The pulmonary and systemic circulations function as two distinct but interconnected vascular systems, each with unique physiological characteristics optimized for their specific roles. While the systemic circulation delivers oxygenated blood at high pressures to tissues throughout the body, the low-pressure pulmonary circulation primarily serves gas exchange, moving oxygen and carbon dioxide between gas (alveoli) and liquid (blood). These fundamental differences in purpose are reflected in their contrasting hemodynamic properties, vascular structures, and responses to physiological stimuli like hypoxia.

### Blood Pressure and Flow Resistance

The most striking difference between pulmonary and systemic circulations is the substantial disparity in operating pressures, even though they handle the same cardiac output. The pulmonary circulation functions as a low-pressure, low-resistance system (Table 1), as is reflected in the pulmonary arteries with thinner walls and less smooth muscle and elastic tissue.

Why are pulmonary pressures so low? The pulmonary and systemic cardiac output are equal, but the work needed to move the blood through the pulmonary circulation is lower. The right ventricle only has to pump blood a short distance to the lungs, and to a height of about 10 cm above the heart (Figure 1). Additionally, the low pressure of the pulmonary systems help protect the thin, delicate alveolar capillaries from damage.

The systemic circulation operates at much higher pressures (Table 1) due to the need to pump blood to the entire body, with more resistance because of thicker vessel walls and higher smooth muscle content in arteries. When compared, the pulmonary vascular resistance (PVR) is approximately one-tenth that of systemic vascular resistance (SVR).

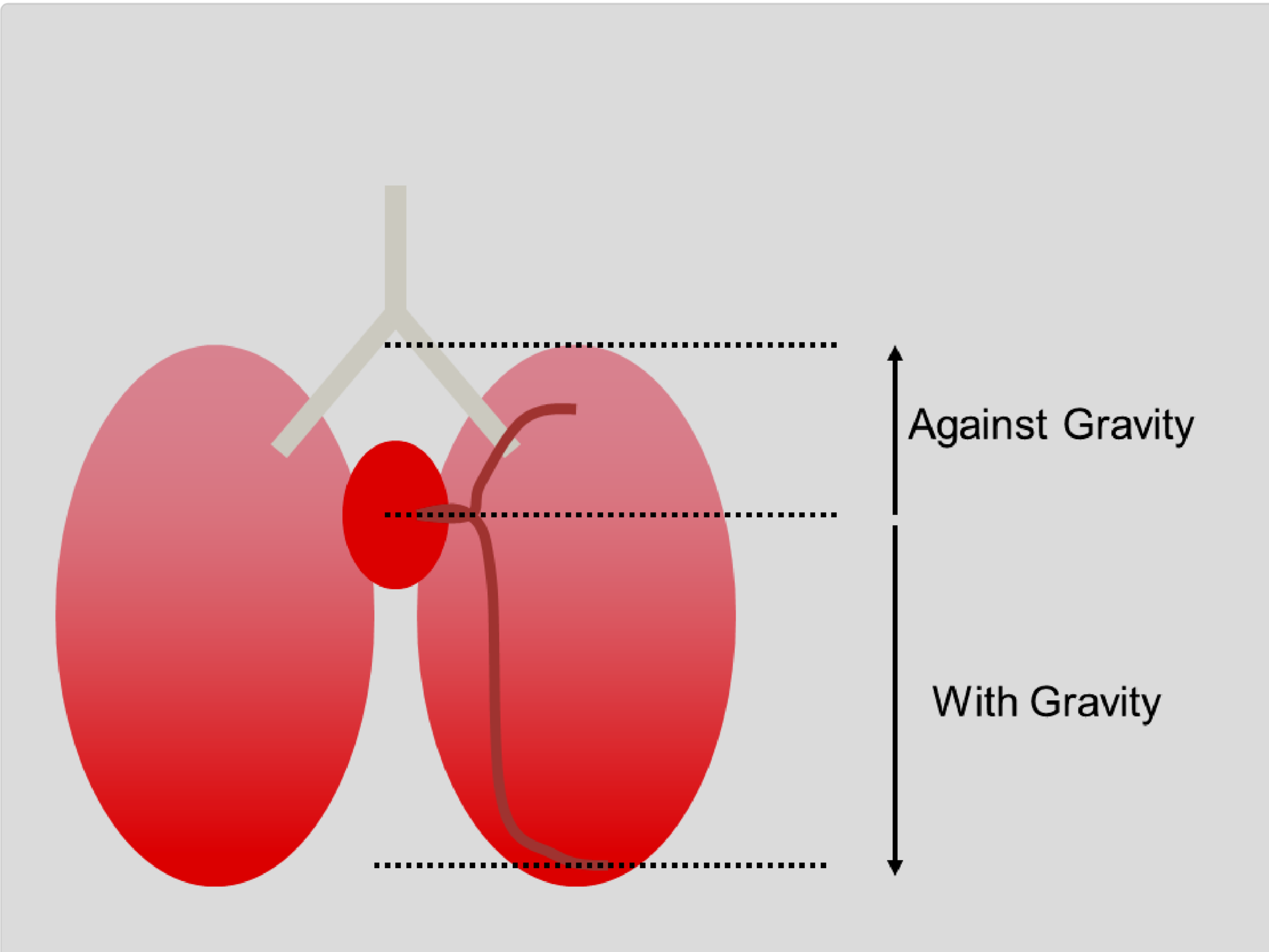


**Figure 1. Distance the heart has to move blood in the lungs**

**Table 1. Comparison of the pressures, flow and resistance of the pulmonary and circulatory systems. Note that cardiac output is equal in both circulations.**

| Pulmonary Circulation | Value | Systemic Circulation | Value |
| --- | --- | --- | --- |

| Pulmonary Circulation | Value | Systemic Circulation | Value |
|---|---|---|---|
| Right Atrium | ~5 mmHg | Left Atrium | ~10 mmHg |
| Right Ventricle | <25 mmHg | Left Ventricle | ~100 mmHg |
| Mean Arterial Pressure | 15 mmHg | Mean Arterial Pressure | 95 mmHg |
| Pressure Gradient | 10 mmHg | Pressure Gradient | 90 mmHg |
| **Cardiac Output** | **5 L/min** | **Cardiac Output** | **5 L/min** |

💵 **Dollar (100¢)** → Left Ventricle (LV) ≈ **100 mmHg**
□ **Quarter (25¢)** → Right Ventricle (RV) ≈ **25 mmHg**
□ **Dime (10¢)** → Left Atrium (LA) ≈ **10 mmHg**
□ **Nickel (5¢)** → Right Atrium (RA) ≈ **5 mmHg**

These structural adaptations have important physiological consequences.

- The thin-walled, compliant pulmonary vessels can accommodate large increases in blood volume with minimal pressure changes. This high compliance allows the pulmonary circulation to function as a reservoir, accommodating variations in blood volume during the cardiac cycle and preventing right ventricular pressure overload.

- The pulmonary capillaries are unique, because with increased flow they can reduce resistance even further via recruitment and distention.
- Additionally, the thin capillary walls are essential for efficient gas exchange, minimizing the diffusion distance for oxygen and carbon dioxide.
- Since the lungs receive 100% of the blood flow, they also play an important role in filtering out small clots through their capillary bed and local thrombolytic processes break down the clots.

## How Is Pulmonary Blood Flow Distributed, and What Are Lung Zones? (LO2)

Pulmonary blood flow is not distributed uniformly throughout the lungs. Instead, it follows predictable patterns influenced by gravity, alveolar blood and air pressures. Understanding these patterns is essential for comprehending normal pulmonary physiology and the pathophysiology of various pulmonary disorders.

**Gravity and Blood Flow**: The right heart generates low pressure, making it challenging to pump blood upward against gravity to the apex (top) of the lungs. However, it is easier for the heart to send blood downward, toward the base of the lungs, when standing. This creates a steep vertical gradient in blood flow (Figure 2).

- **Base**: Blood flow is highest at the base of the lungs because gravity increases hydrostatic pressure, making it easier for blood to be pumped into these areas.

- **Apex**: Blood flow is lowest at the apex, as gravity reduces perfusion, making it harder for blood to reach this region.

Body position significantly affects this distribution. In the supine position (lying down), the vertical height of the lungs is reduced, diminishing gravitational effects. Blood flow becomes more uniform, with slightly greater perfusion to posterior (dependent) regions. In the lateral decubitus position (lying on their side), the dependent (lowermost) lung receives greater blood flow than the non-dependent lung.

**Pulmonary Blood Flow Distribution and West's Lung Zones.** West's lung zones divide the upright lung into three regions based on the relationship among pulmonary arterial pressure ($P_a$), venous pressure ($P_V$), and alveolar pressure ($P_A$). Gravity creates a hydrostatic gradient: arterial and venous pressures are lowest at the apex and highest at the base, while alveolar pressure is roughly uniform throughout. The interplay of these three pressures determines whether capillaries are open, partially compressed, or fully perfused, and therefore how much blood flow each region receives (Figure 2).

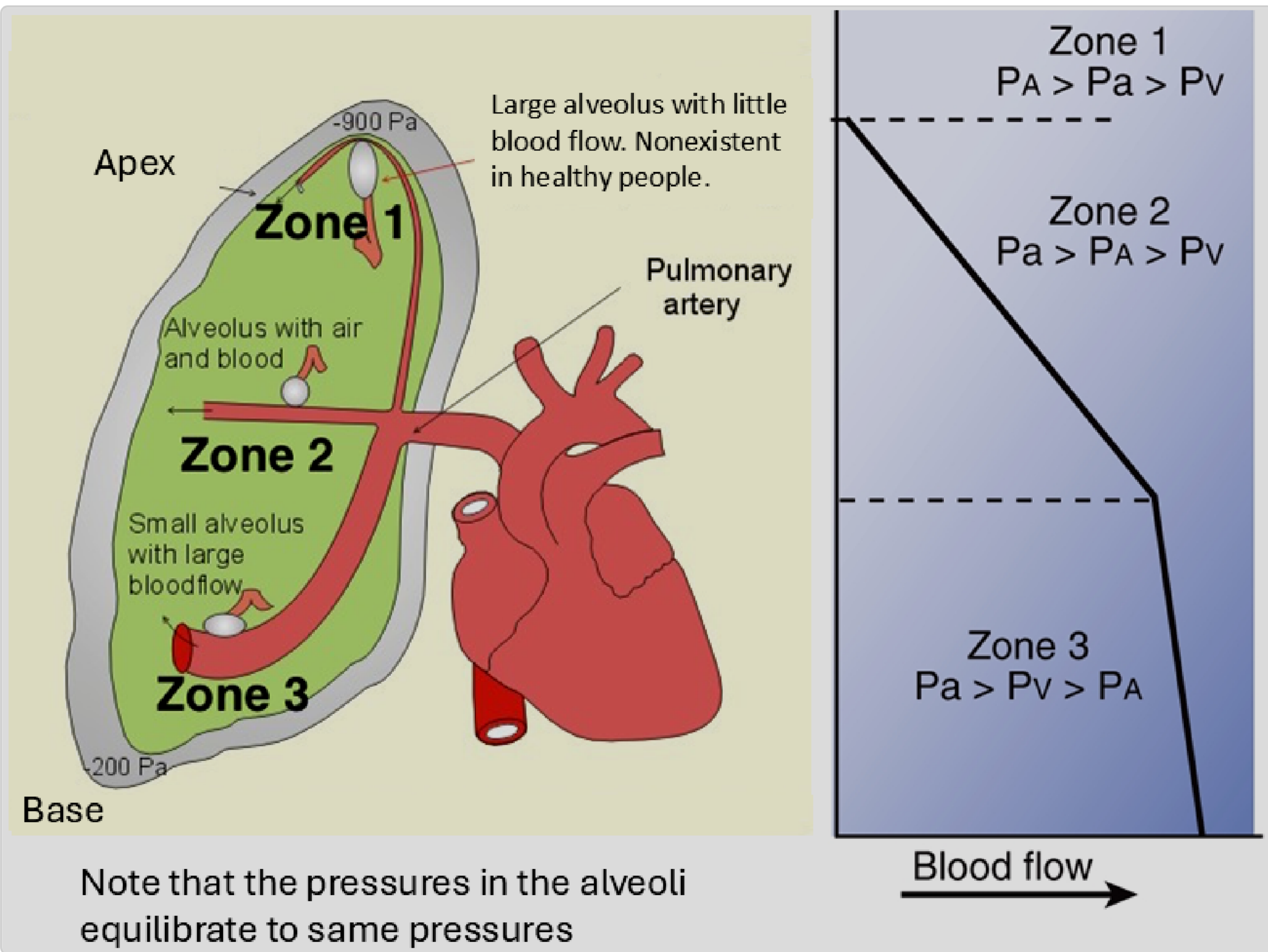


**Figure 2. West zones in the upright lung- demonstrating how regional blood flow is affected by Alveolar, and arterial and venous pressures.**

**Zone 1:** When present, occurs primarily in the **apex.** The alveolar pressure exceeds pulmonary arterial pressure, causing vessel collapse, "crushing" the vessels, and eliminating flow. These alveoli are ventilated but not perfused, creating **dead space**. (Figure 2). This zone is generally absent in healthy individuals but may develop in conditions of: positive pressure ventilation, hypovolemia, or shock when pulmonary arterial pressure decreases. (Zone 1 Apex: $P_A > P_a > P_V$).

**CLINICAL CORRELATION**

**Latent tuberculosis** affects zone 1, where it tends to reactivate in the apex of the lungs, where the $\dot{V}/\dot{Q}$ ratio is high. The increased ventilation in this region creates an oxygen-rich environment, allowing *Mycobacterium tuberculosis*, an organism that thrives in oxygen, to flourish.

This represents an observed pattern. Aerobic microorganisms prefer the apex regions of the lung with higher $PO_2$ while anaerobic microorganisms proliferate better in the base of the lung.

**Zone 2:** In the middle region of the lung, **a**rterial pressure is high enough to open the upstream end of capillaries. However, alveolar pressure still exceeds venous pressure and partially compresses the downstream end of the capillary. Because of this, blood flow is determined by the pressure gradient along the length of the capillary (Figure 2). Moving down through Zone 2, the rising hydrostatic pressure due to gravitational effects, progressively opens more of each capillary, so blood flow increases steadily toward the base. (Zone 2 Middle: $P_a > P_A > P_V$).

**Zone 3:** At the base of the lung, both arterial and venous pressures exceed alveolar pressure, so capillaries are fully open. Blood flow here is determined by the arterial-venous pressure difference ($P_a$ - $P_V$), following standard hemodynamic principles. Increasing hydrostatic pressure toward the base further augment flow by recruitment and distention of additional capillary beds. (Zone 3 Base: $P_a > P_V > P_A$).

**Net result:** The lung operates as a mixture of these zones. The net result is that perfusion and ventilation are both greater in the base of the lung than in the apex and we only see limited Zone 1 flow in disease states.

Pulmonary arterial ($P_a$) pressures keep the capillaries open and maintain flow through the capillary beds. Pulmonary venous ($P_V$) pressures determine the pressure gradient for flow leaving the pulmonary capillaries. The alveolar pressure ($P_A$) is proportional to its volume and compresses the capillaries when maximally inspiring.

## How Does Pulmonary Vascular Resistance (PVR) Change With Cardiac Output, Arterial Pressure, and Lung Volume (LO3)?

Like systemic vascular resistance (SVR), the **pulmonary vascular resistance** (PVR) is the total resistance to blood flow in the pulmonary circulation as primarily determined by the regulated flow through arterioles and capillary beds.

Also like SVR, PVR is dependent on changes with arterial compliance, cardiac output, and arterial pressure. In the lungs, we also see that changes in resistance are also affected by changes in lung volume.

**Pulmonary Arterial Compliance.** The pulmonary arterial circulation demonstrates high compliance (expanding with stretch), because the thin, compliant pulmonary arterial wall expands, allowing it to accommodate significant changes in blood volume with minimal pressure changes. This prevents right ventricular pressure overload and maintaining efficient gas exchange.

**Capillary recruitment and distension.** The capillaries are also able to lower resistance with increased cardiac output, with the following capillary bed changes occurring:

- **Recruitment:** Previously unperfused capillaries are pushed open, effectively increasing the cross-sectional area of the pulmonary vascular bed.

- **Distension:** already perfused vessels expand (distension) due to their high compliance (Figure 3).

PVR is not static but responds dynamically to various physiological conditions.

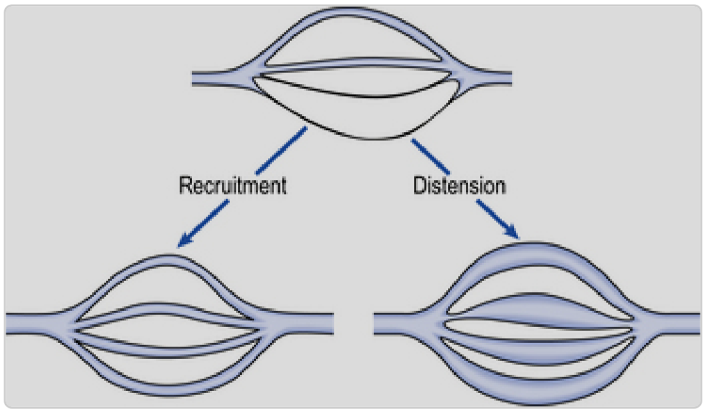


**Figure 3. recruitment and distension of pulmonary blood vessels.**

According to Poiseuille's law, resistance is inversely proportional to the fourth power of the radius ($R \propto 1/r^4$), so even small increases in vessel diameter significantly reduce resistance. These mechanisms allow the pulmonary circulation to handle increased cardiac output without significant pressure increases.

**CLINICAL CORRELATION**

**Distension and recruitment:** Through recruitment and distension of pulmonary vessels, the pulmonary circulation can accommodate up to a five-fold increase in cardiac output during maximal exercise, with only a slight rise in pulmonary arterial pressure. This keeps pressures below 25 mmHg under normal conditions, protecting the delicate alveoli from injury and edema.

However, this adaptive capacity has limits. In conditions of chronically elevated cardiac output or in pulmonary vascular disease, recruitment and distension mechanisms may be exhausted, leading to pulmonary hypertension.

**Lung Volume:** Changes in lung volume also affect PVR, with a U-shaped curve relationship, with PVR reaching its minimum at functional residual capacity (FRC) and increasing at both lower and higher lung volumes (Figure 4). Pulmonary vascular resistance is high both when the lung is maximally deflated (at residual volume) or maximally inflated (at total lung capacity), and it is lowest at the end of a normal expiration (i.e., at functional residual capacity; FRC).

- At Residual Volume, pressure is increased as pleural pressures increase during a forced expiration.
- As lung volume approaches Functional Residual Capacity (FRC), pulmonary pressures reduce and capillaries see increased flow as compression is relieved, resistance decreases.
- With further inspiration, the alveoli continue to expand, exerting force against the adjacent capillaries, restricting blood flow and increasing resistance.

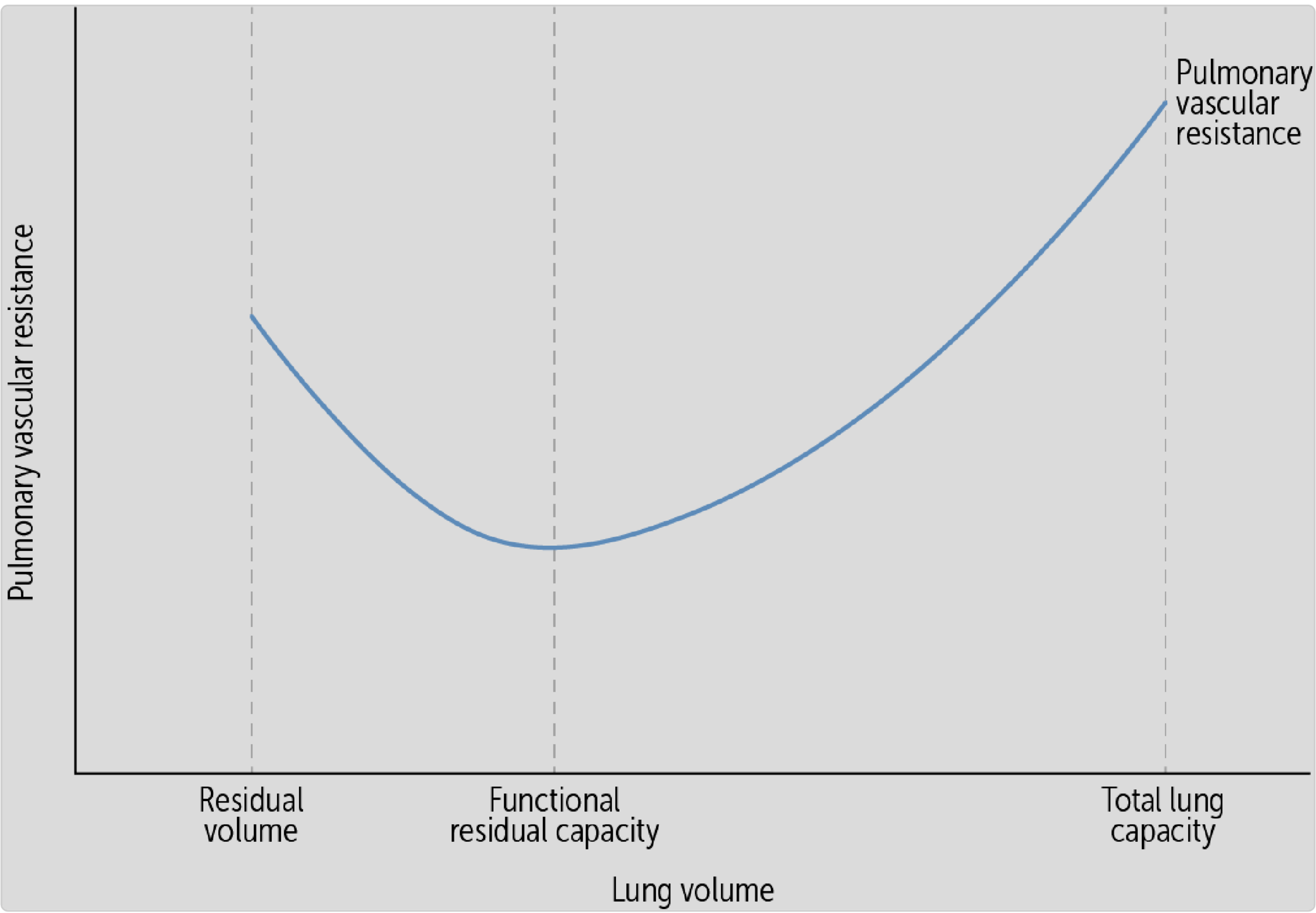


**Figure 4 Pulmonary vascular resistance vs lung capacity**

PVR decreases due to recruitment and distension of pulmonary vessels (to prevent edema).

## What Are the Effects of Hypoxic Pulmonary Vasoconstriction (HPV) (LO4)?

**Hypoxic pulmonary vasoconstriction (HPV)** is a distinctive feature of the pulmonary circulation. Unlike systemic vessels that dilate in response to hypoxia to increase blood flow to oxygen-deprived tissues, pulmonary vessels constrict. This change in pulmonary vascular resistance diverts blood away from poorly oxygenated areas of the lung (Figure 5). This seemingly paradoxical response serves a crucial physiological purpose: optimizing the match between ventilation and perfusion to increase systemic oxygen.

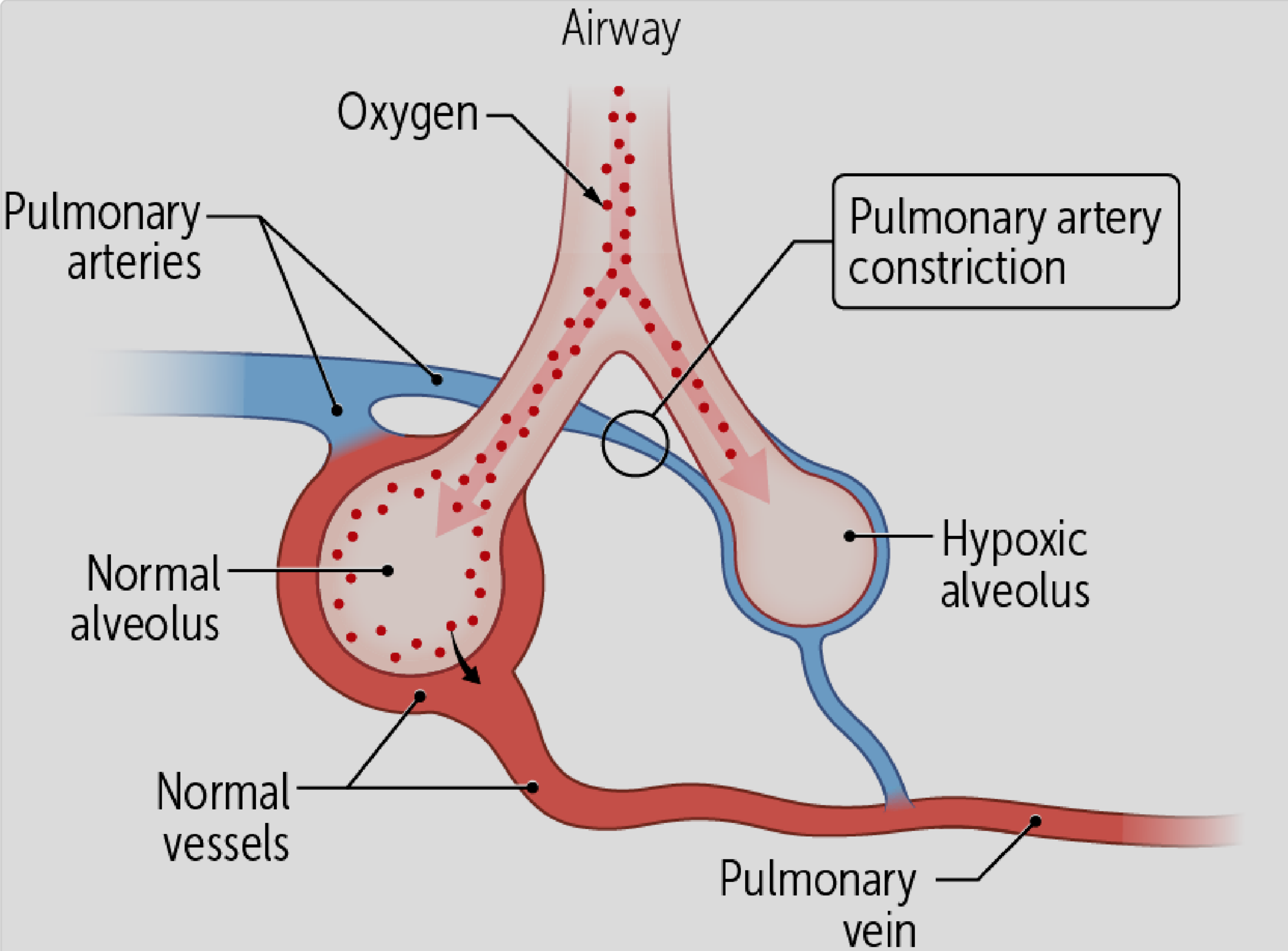


**Figure 5. Hypoxic pulmonary vasoconstriction- the alveoli on the right has reduced airflow, and resultant hypoxic vasoconstriction (blue colored arteries).**

**How does hypoxic pulmonary vasoconstriction (HPV) work?** HPV is primarily triggered by alveolar hypoxia (low $P_AO_2$) (Figure 6), rather than arterial hypoxemia (low $P_aO_2$), though both can contribute. When an area of the lungs has low alveolar oxygen partial pressure (low tension), the capillaries surrounding the low-oxygen alveoli also develop low-oxygen content (hypoxemia). While several factors, such as oxygen-sensitive potassium channels, nitric oxide (NO), and endothelin-1, have been suggested to play a role in the HPV mechanism, there is no clear consensus on the exact molecular mechanism behind this response. Regardless, the smooth muscle cells depolarize and contract, decreasing the vessel radius (**vasoconstriction**) and increasing the resistance in these vessels.

**What is the result of this local increase in PVR?** Blood flow is decreased to this region, and shunted from the hypoxic lung area to other areas of the lung that have better oxygen ventilation (Figure 5). The response begins within seconds of exposure to hypoxia, reaches maximum effect within minutes, and is fully reversible upon restoration of normal oxygen levels. This shunting maximizes overall blood oxygenation.

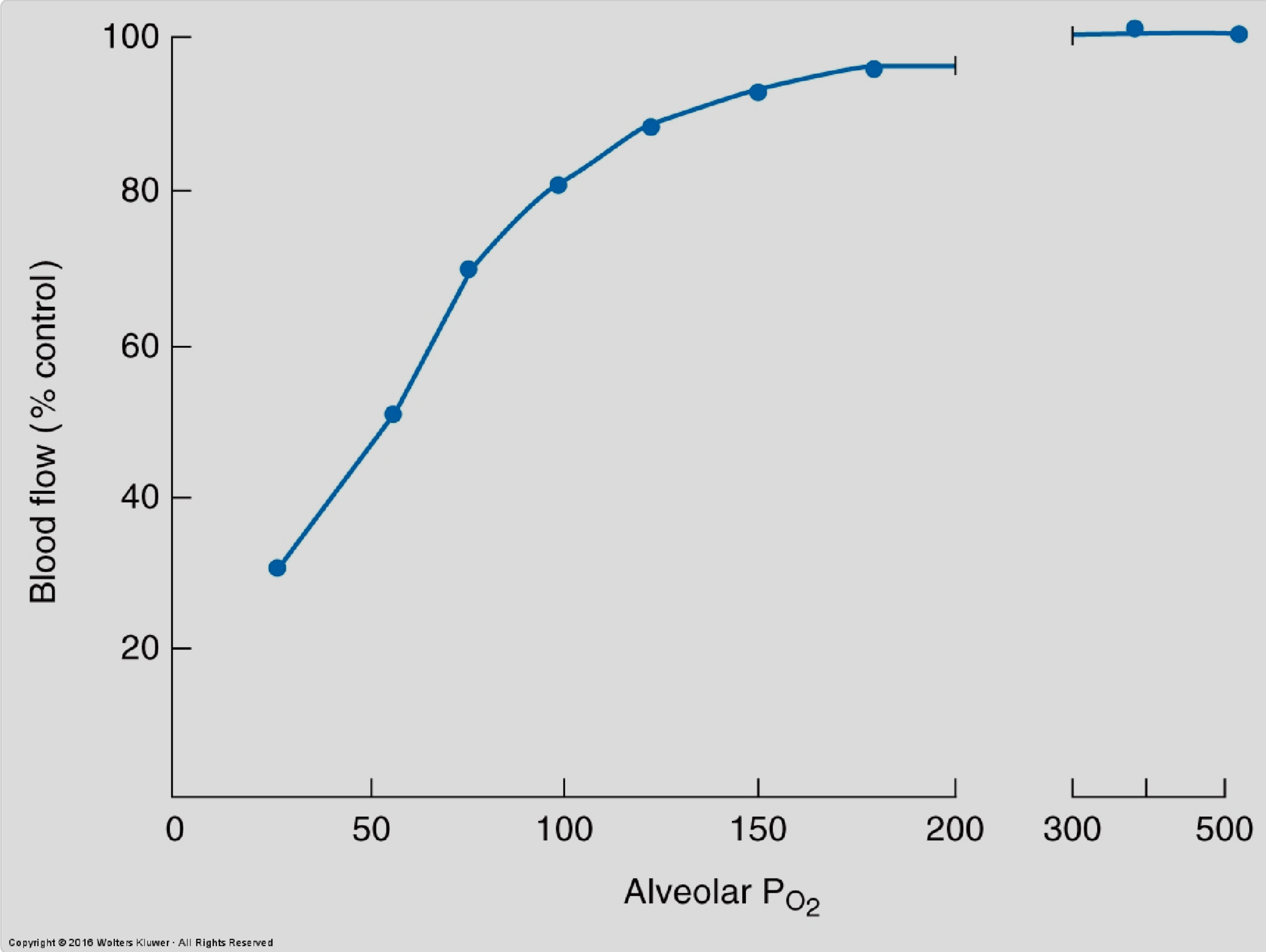


**Figure 6. Hypoxic vasoconstriction: lung blood flow changes with alveolar oxygen levels ($PaO_2$)**

HPV can occur in isolated lung segments, lobes, or throughout both lungs, depending on the distribution of hypoxia. In **localized lung diseases** such as pneumonia, atelectasis, or bronchial obstruction, HPV reduces blood flow only to those affected regions, minimizing the impact on systemic oxygenation. Without this response, these conditions would cause more severe arterial hypoxemia due to increased shunting of deoxygenated blood.

**CLINICAL CORRELATION**

**When oxygen levels drop throughout the lungs** (like at high altitude or with poor breathing), all the lung's blood vessels constrict. While the HPV response normally helps redirect blood to better-oxygenated areas, it is maladaptive when the entire lung lacks oxygen.

**The result:** The right side of the heart must work harder to pump blood through these constricted vessels.  Over time, this pulmonary hypertension can lead to right ventricular hypertrophy, a condition called cor pulmonale. Additionally, the high pulmonary arterial pressure may increase capillary hydrostatic pressure, potentially leading to pulmonary edema.

**High-altitude pulmonary edema (HAPE)** combines both problems: blood vessel constriction raises pulmonary arterial pressures while uneven constriction damages capillaries, making them leak fluid into the lungs (pulmonary edema).

Pulmonary vessels constrict in response to hypoxia and redirects it to well-ventilated regions of the lung. (Note: systemic vessels dilate with hypoxia).

## CLINICAL CORRELATION

**Pulmonary Hypertension**: Chronic hypoxia, as seen in COPD, leads to sustained HPV, increasing pulmonary arterial pressure and causing right ventricular hypertrophy (cor pulmonale).

## ACTIVE LEARNING QUESTION

What is the primary function of hypoxic pulmonary vasoconstriction (HPV)?

**A.** Maximizing blood oxygenation

**B.** Reducing arterial hypoxemia

**C.** Preventing lung damage

**D.** Improving ventilation

**E.** Enhancing systemic circulation

**Explanation** (requires correct answer)

## What Are the Mechanisms and Causes of Pulmonary Edema (LO5)?

**Pulmonary edema** is abnormal fluid buildup in the lungs that occurs when the balance of forces controlling fluid movement across capillary walls is disrupted. Under normal conditions, Starling's forces maintain equilibrium: hydrostatic pressure pushes fluid out of capillaries while colloid osmotic (oncotic) pressure from proteins pulls it back in (Figure 7). In the lungs, air pressure pushes fluid toward capillaries while surface tension pulls it into alveoli (though surfactant reduces this effect, Figure 7). Any excess fluid is cleared by lymphatic drainage.

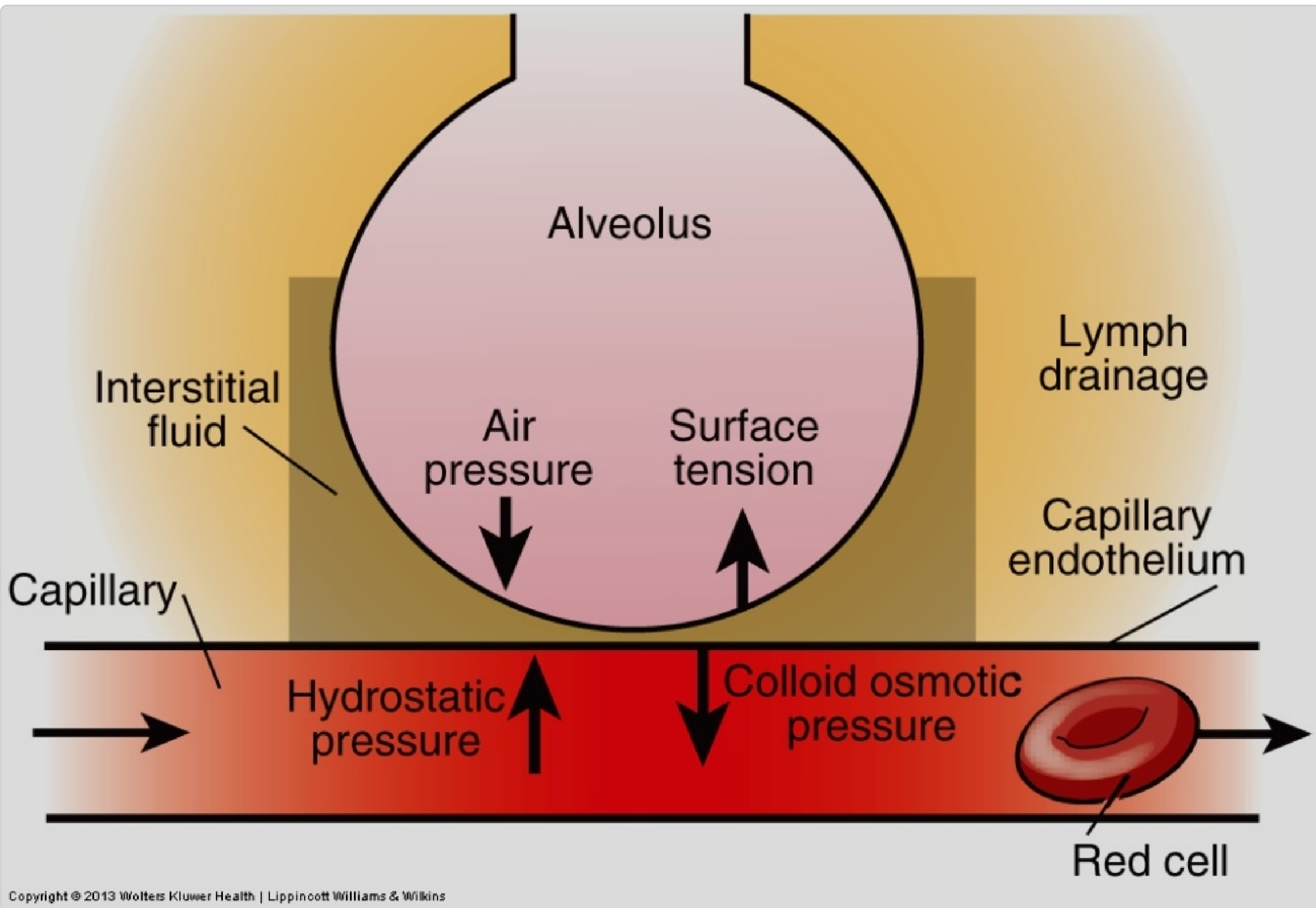


**Figure 7. Starlings forces in the lungs which control fluid movement between the alveoli, interstitium and alveolar capillaries.**

Pulmonary edema can be classified into two major types, depending on the causation:

- **Hydrostatic (cardiogenic) edema** results from increased pressure in pulmonary capillaries occurs when capillary pressure rises above 25 mmHg (normal is 7-12 mmHg), it overcomes the colloid osmotic pressure of 25-28 mmHg, causing net fluid movement into lung tissue (see pressures in Figure 7). This commonly occurs in left heart failure when blood backs up into the lungs.
- **Permeability (non-cardiogenic) edema** occurs when capillary walls become damaged and leaky, allowing protein-rich fluid to escape. The leaked proteins worsen the edema by increasing interstitial colloid osmotic pressure, reducing the oncotic gradient opposing capillary filtration. In addition, the protein-rich edema fluid disrupts surfactant function, increasing surface tension and promoting alveolar collapse. Both of these changes draw even more fluid from the capillaries, worsening edema.

Pulmonary edema progresses in stages as the lung's protective mechanisms are overwhelmed.

1. Initially, lymphatic flow can increase to clear excess fluid. However, when this capacity is exceeded, **interstitial edema** develops around blood vessels (perivascular) and bronchial spaces (airways) (Figure 8, #1).
2. While fluids rarely accumulate in healthy alveoli, they do so in later stages of edema (Figure 8, #2).

The resulting **alveolar edema** severely impairs gas exchange. The cause of alveolar edema is unknown but may be due to the excess fluid exceeding the lymphatic clearance rate.

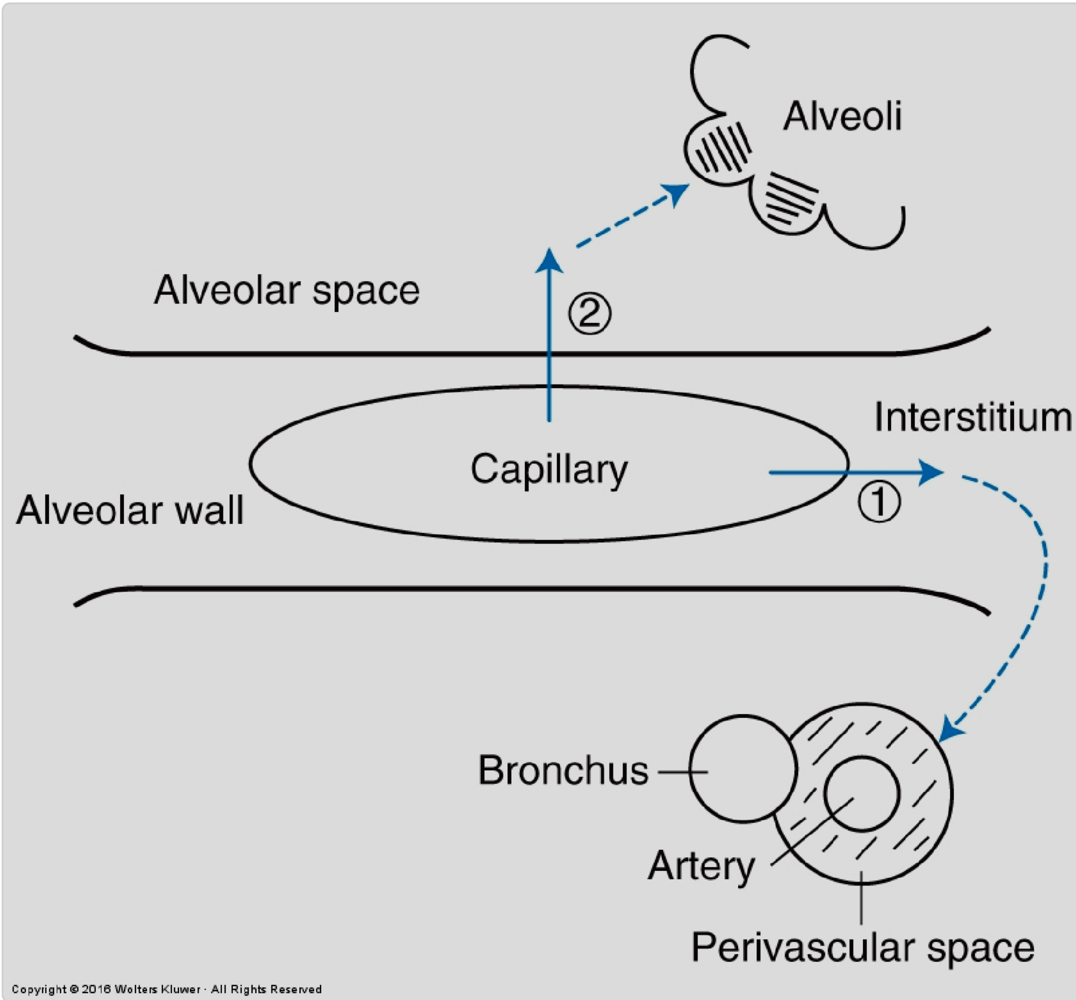


**Figure 8. Fluid pathways from pulmonary capillaries: 1. Interstitium where edematous fluid is removed by lymphatics in perivascular space. 2. If the edema is excessive, then fluid can accumulate in the alveoli themselves, and affect gas exchange.**

The different causes of edema, related to the factors in Figure 7 and Figure 8 are shown in Table 1.

**Table 2. Summary of causes of pulmonary edema**

| Causes of Edema | Effect |
|---|---|
| **Increased Hydrostatic Pressure** | Seen in left-sided heart failure, leading to fluid extravasation (leakage) into the alveoli. |
| **Increased Vessel Permeability** | Due to inflammation or toxic substances, increasing fluid leakage into the interstitium and alveoli. |
| **Impaired Lymphatic Drainage** | There is a net movement of fluid from the vascular to the extracellular space, about 2-4 L/day. This is volume is cleared by the lymphatic system. Tumors or lymphangitis can obstruct lymphatics, leading to fluid accumulation. |
| **Decreased Oncotic Pressure** | Oncotic pressures can be reduced by 1) over-transfusion or saline perfusion, or 2) liver disease with loss of albumin. Both of these result in less albumin, and oncotic pressure and fluid leakage from the capillaries. |
| **Lack of surfactant** | Surfactant reduces this edema causing effect of surface tension. Surface tension pulls on the alveolus and decreases the hydrostatic pressure in the extracellular space — causing water to be sucked into that space. |

In LV heart failure, the blood pressure backs up into the pulmonary circulation and affects the right ventricle (RV). This increased hydrostatic pressure in pulmonary capillaries leads to fluid leakage into alveoli.

**CASE CONNECTION** BACK TO INTRODUCTION

Thinking back to CH's case, his pulmonary circulation responds to hypoxemia with hypoxic vasoconstriction (HPV), while his systemic circulation responds with vasodilation. This HPV helps optimize ventilation-perfusion matching in the lungs by diverting blood from poorly ventilated areas to better-ventilated regions.

However, as COPD progresses, the chronic hypoxemia leads to widespread HPV, contributing to pulmonary hypertension. This increased pressure, combined with potential right heart failure (cor pulmonale), raises hydrostatic pressure in pulmonary capillaries, predisposing CH to pulmonary edema. This underlies his dyspnea (shortness of breath) and low $O_2$ sats. Additionally, inflammatory processes in COPD may damage his alveolar-capillary membrane, increasing permeability and further contributing to edema risk.

# Summary

- Pulmonary circulation operates at lower pressures than systemic circulation to protect the fragile alveolar capillaries.
- Pulmonary blood flow distribution in the upright lung is influenced by gravity, with the highest flow at the base of the lungs and the lowest at the apex.
- This blood flow distribution in the upright lung follows West's lung zones: Zone 1 (minimal flow, absent in healthy individuals), Zone 2 (flow

determined by $P_a - P_A$), and Zone 3 (normal flow through the base).

- Pulmonary vascular resistance decreases with increasing **cardiac output** through capillary recruitment and distension mechanisms and compliant arteries, and varies with **lung volume** (lowest at FRC, highest at residual volume and total lung capacity).
- Hypoxic pulmonary vasoconstriction (HPV) is a unique response where pulmonary vessels constrict in hypoxic regions (opposite to systemic vasodilation) by improving  blood oxygenation by diverting blood to better-ventilated lung areas.
- Pulmonary edema can be caused by increased hydrostatic pressure, increased alveolar pressure, increased permeability, impaired lymphatic drainage, hemodilution, or lack of surfactant.

## Review Questions

1. What is the most significant difference between pulmonary and systemic circulations regarding vascular resistance, under normal conditions?

**A.** Pulmonary resistance is lower.

**B.** Pulmonary resistance is higher

**C.** Systemic resistance is unaffected by gravity

**D.** Systemic resistance increases with hypoxia.

**Explanation** (requires correct answer)

**2.** A healthy 25-year-old begins exercising vigorously. How does pulmonary vascular resistance (PRV) most likely change as cardiac output increases from 5 L/min to 25 L/min?

**A.** Remains unchanged

**B.** Decreases significantly

**C.** Increases significantly

**D.** Decreases slightly

**E.** Increases slightly

**Explanation** **(requires correct answer)**

**3.** How does hypoxic pulmonary vasoconstriction affect blood flow in hypoxic lung regions?

**A.** increases by dilation

**B.** decreases due to parasympathetic stimulation

**C.** decreases by vasoconstriction

**D.** remains unchanged

**Explanation** **(requires correct answer)**

A. Alveolar oxygen concentration

B. Gravity and hydrostatic pressure gradients

C. Airway diameter

D. Lung compliance

Explanation (requires correct answer)

5. Pulmonary vascular resistance is lowest at which lung volume?

A. Residual volume

B. Functional residual capacity

C. Total lung capacity

D. Tidal volume

**Explanation** (requires correct answer)

## How would you rate this Brick?

## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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04 Discuss the effects of hypoxic pulmonary vasoconstriction on blood flow distribution in the lungs.

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Objective Confidence: 0/5